Brazilian E-Commerce Public Dataset

6. Feature Engineering Report

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1. Scope

- Feature engineering was performed on order_level_clean.csv and item_level_clean.csv to produce model-ready inputs and aggregated analytical tables.
- Four output artefacts created: order_level_features.csv, customer_features.csv, seller_features.csv, and product_features.csv.
- All features are interpretable and designed to support downstream tasks: churn prediction, demand forecasting, seller scoring, and customer segmentation.

2. Order-Level Features

2.1 Delivery and Timing

Feature	Formula	Type	Purpose
delivery_days	order_delivered_customer_date - order_purchase_timestamp (days)	numeric	Actual end-to-end delivery duration
delivery_vs_estimate_days	order_delivered_customer_date - order_estimated_delivery_date	numeric	Lateness measure (negative = arrived early)
is_late	1 if delivery_vs_estimate_days > 0 else 0	binary	Lateness flag for satisfaction modelling
purchase_year	order_purchase_timestamp.dt.year	int	Year dimension for temporal analysis
purchase_month	order_purchase_timestamp.dt.month	int	Month dimension for seasonality
purchase_dayofweek	order_purchase_timestamp.dt.dayofweek	int	Day-of-week pattern (0=Mon, 6=Sun)

- First 5 rows show delivery_vs_estimate_days ranging from -16 to -28 days -- the platform consistently over-estimates delivery time, protecting review scores.

2.2 Payment Features

Feature	Formula	Type	Purpose
uses_installments	1 if max_installments > 1 else 0	binary	Flags instalment-based buyers
order_value_segment	pd.cut on total_payment_value	categorical	Low (<R$50) / Medium (R$50-150) / High (R$150-500) / Premium (>R$500)

3. Customer-Level Features

- Aggregated from order_level_features.csv at customer_unique_id grain -- 9,956 unique customers.

Feature	Aggregation	Purpose
total_spent	SUM(total_payment_value)	Total lifetime value of customer
n_orders	NUNIQUE(order_id)	Number of orders placed
avg_order_value	MEAN(total_payment_value)	Average spend per transaction
avg_review_given	MEAN(avg_review_score)	Customer average satisfaction rating
avg_delivery_days	MEAN(delivery_days)	Average delivery experience
total_installments_used	SUM(uses_installments)	Frequency of instalment payment usage
is_repeat_customer	1 if n_orders > 1 else 0	Repeat buyer binary flag
customer_tier	pd.cut on total_spent	Bronze (<R$100) / Silver (R$100-300) / Gold (R$300-1,000) / Platinum (>R$1,000)

- Output: customer_features.csv -- 9,956 rows x 9 columns saved to data/processed/.

- Nearly all customers land in Bronze or Silver tiers, reflecting a single low-value purchase base with minimal repeat activity.

4. Seller-Level Features

- Aggregated from item_level_clean.csv at seller_id grain -- 1,654 unique sellers.

Feature	Aggregation	Purpose
seller_total_revenue	SUM(item_revenue)	Total revenue generated by seller
seller_n_items	COUNT(order_item_id)	Total items sold
seller_n_orders	NUNIQUE(order_id)	Distinct orders fulfilled
seller_avg_price	MEAN(price)	Average price point -- premium vs. volume indicator
seller_state	FIRST(seller_state)	Seller geographic base

- Output: seller_features.csv -- 1,654 rows x 6 columns saved to data/processed/.
- Top 3 sellers (all SP-based) each generated ~R$24,000. One used 201 items at avg R$106, another 57 items at avg R$406 -- contrasting volume vs. premium strategies.

5. Product-Level Features

- Aggregated from item_level_clean.csv at product_id grain -- 6,747 unique products.

Feature	Aggregation	Purpose
product_items_sold	COUNT(order_item_id)	Total units sold -- popularity metric
product_total_revenue	SUM(item_revenue)	Total revenue contribution
product_avg_price	MEAN(price)	Average selling price
product_category	FIRST(category)	Category label for grouping

- Output: product_features.csv -- 6,747 rows x 5 columns saved to reports/.
- Top product by units sold: a garden_tools item with 56 units and R$4,043 revenue.

6. Engineered Feature Summary

Output Table	Rows	New Features	Use Case
order_level_features.csv	10,000	8 (delivery, payment, time)	Delivery modelling, trend analysis
customer_features.csv	9,956	8 (RFM-style + tier)	Segmentation, churn prediction
seller_features.csv	1,654	5 (revenue, volume, geo)	Seller scoring, recommendation
product_features.csv	6,747	4 (sales, revenue, category)	Demand forecasting, catalogue mgmt